

Measuring error

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Take-aways from last class activity

- ▶ Even if the model is correctly specified, prediction performance on the **training** data is a biased estimate of prediction performance on **new** data
- ▶ The bias gets worse as the number of explanatory variables in the model increases!

Bias of the training error

Let $\mathcal{T} = \{(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)\}$ be a training set, and let $\hat{h}$ be a model fit on $\mathcal{T}$ by maximizing the likelihood.

Two big results (under regularity conditions):

Suppose we use $-2 \log$ likelihood as loss function. Then:

Theorem 1: If model is correctly specified,

$$\mathcal{R} \approx \mathbb{E}_{\mathcal{T}} [\mathcal{R}_{\text{train}}(\hat{h})] + \underbrace{\left(\frac{2p}{n} \right)}_{\text{bias correction (depends on } p \text{ \& } n!)} \leftarrow$$

(expected test error)

Theorem 2: $\mathcal{R}_{\text{Loocv}} = \frac{1}{n} \sum_{i=1}^n L(Y_i, \hat{h}_{-i}(x_i))$

Using $-2 \log$ likelihood as loss:

$$\mathcal{R} \approx \mathbb{E}_{\mathcal{T}} [\mathcal{R}_{\text{Loocv}}]$$

(expected test error)

(model does not have to be correctly specified!)

Deriving bias of the training error (for simplicity, ignoring x)

Let $\mathbf{y} = (Y_1, \dots, Y_n) \stackrel{iid}{\sim} g$ for some unknown distribution g be our **training set**, and let $\mathbf{y}^* = (Y_1^*, \dots, Y_n^*) \stackrel{iid}{\sim} g$ be another sample from g , independent of $\mathbf{y}$.

g is unknown ; assume (may be wrong!) that
 $g \in \{f_\theta : \theta \in \Theta \subseteq \mathbb{R}^p\}$ (p -dimensional parametric model)

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmax}} \log f_\theta(\mathbf{y}) \quad (\text{MLE!})$$

$$\text{score: } u(\theta) = \frac{\partial}{\partial \theta} \log f_\theta(\mathbf{y}) \quad \text{Hessian: } H(\theta) = \frac{\partial^2}{\partial \theta^2} \log f_\theta(\mathbf{y})$$

$$\hat{\theta} \xrightarrow{P} \theta_0 \quad \theta_0 \text{ solves } \mathbb{E}_{\mathbf{y}} \left[\frac{\partial}{\partial \theta} \log f_\theta(\mathbf{y}) \right] = 0$$

$$A = -\mathbb{E}_{\mathbf{y}} [H(\theta_0)] \quad B = \operatorname{Var}_{\mathbf{y}}(u(\theta_0)) = \mathbb{E}_{\mathbf{y}} [u(\theta_0) u(\theta_0)^T]$$

Error quantities

Let $\mathbf{y} = (Y_1, \dots, Y_n) \stackrel{iid}{\sim} g$ for some unknown distribution g be our **training set**, and let $\mathbf{y}^* = (Y_1^*, \dots, Y_n^*) \stackrel{iid}{\sim} g$ be another sample from g , independent of $\mathbf{y}$.

$$\mathcal{R}_{\text{train}}(\hat{\theta}) = -\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}) = -\frac{2}{n} \sum_{i=1}^n \log f_{\hat{\theta}}(y_i)$$

$$\mathcal{R}_r(\hat{\theta}) = \mathbb{E}_{\mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}^*) \mid \mathbf{y} \right]$$

$$\mathcal{R} = \mathbb{E}_{\mathbf{y}} [\mathcal{R}_r(\hat{\theta})] = \mathbb{E}_{\mathbf{y}, \mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}^*) \right]$$

Q: $\mathbb{E}_{\mathbf{y}} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}) \right]$ vs $\mathbb{E}_{\mathbf{y}, \mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}^*) \right]$

depends on $\mathbf{y}$ (training) *↑* *↑* *applied to $\mathbf{y}^*$ (test)*

Idea

know $\hat{\theta} \rightarrow \theta_0$. Connect w/ θ_0 ?

$$\mathcal{R}_{\text{train}}(\hat{\theta}) = -\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y})$$

?

$$\mathcal{R}_{\mathcal{T}}(\hat{\theta}) = \mathbb{E}_{\mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}^*) \middle| \mathbf{y} \right]$$

$$\mathcal{R} = \mathbb{E}_{\mathbf{y}}[\mathcal{R}_{\mathcal{T}}(\hat{\theta})] = \mathbb{E}_{\mathbf{y}, \mathbf{y}^*} \left[-\frac{2}{n} \log f_{\hat{\theta}}(\mathbf{y}^*) \right]$$

$$\mathbb{E}_{\mathbf{y}} [-2 \log f_{\theta_0}(\mathbf{y})] \approx \mathbb{E}_{\mathbf{y}} [-2 \log f_{\hat{\theta}}(\mathbf{y})] + \underbrace{\text{tr}(\mathbf{A}^{-1}\mathbf{B})}$$

$$\mathbb{E}_{\mathbf{y}, \mathbf{y}^*} [-2 \log f_{\hat{\theta}}(\mathbf{y}^*)] \approx \mathbb{E}_{\mathbf{y}} [-2 \log f_{\theta_0}(\mathbf{y})] + \underbrace{\text{tr}(\mathbf{A}^{-1}\mathbf{B})}$$

$$\Rightarrow \mathbb{E}_{\mathbf{y}, \mathbf{y}^*} [-2 \log f_{\hat{\theta}}(\mathbf{y}^*)] \approx \mathbb{E}_{\mathbf{y}} [-2 \log f_{\hat{\theta}}(\mathbf{y})] + 2 \text{tr}(\mathbf{A}^{-1}\mathbf{B})$$

$$\mathcal{R} \approx \mathbb{E}_{\mathbf{y}} [\mathcal{R}_{\text{train}}(\hat{\theta})] + \frac{2}{n} \text{tr}(\mathbf{A}^{-1}\mathbf{B})$$

if $\mathbf{A} = \mathbf{B}$
(correctly specified)

$$\mathcal{R} \approx \mathbb{E}_{\mathbf{y}} [\mathcal{R}_{\text{train}}(\hat{\theta})] + \frac{2p}{n}$$

Lemma 1 proof

Taylor expansion of $\log f_{\theta_0}(y)$ around $\hat{\theta}$ ($\hat{\theta} \xrightarrow{p} \theta_0$)

$$\log f_{\theta_0}(y) \approx \log f_{\hat{\theta}}(y) + (\theta_0 - \hat{\theta})^T u(\hat{\theta}) + \frac{1}{2} (\theta_0 - \hat{\theta})^T H(\hat{\theta}) (\theta_0 - \hat{\theta})$$

$$u(\hat{\theta}) = 0$$

(MLE)

Taylor expansion of u :

$$\Rightarrow \underbrace{u(\hat{\theta})}_0 \approx u(\theta_0) + H(\theta_0)(\hat{\theta} - \theta_0)$$

$$\Rightarrow -(\hat{\theta} - \theta_0) \approx H^{-1}(\theta_0) u(\theta_0)$$

$$\Rightarrow \theta_0 - \hat{\theta} \approx H^{-1}(\theta_0) u(\theta_0)$$

$$\hookrightarrow \log f_{\theta_0}(y) \approx \log f_{\hat{\theta}}(y) + \frac{1}{2} u(\theta_0)^T H^{-1}(\theta_0) H(\hat{\theta}) H^{-1}(\theta_0) u(\theta_0)$$

Lemma 1 proof

$$\begin{aligned}\log f_{\theta_0}(y) &\approx \log f_{\hat{\theta}}(y) + \frac{1}{2} u(\theta_0)^T H^{-1}(\theta_0) \underbrace{H(\hat{\theta})}_{\approx H(\theta_0)} H^{-1}(\theta_0) u(\theta_0) \\ &\approx \log f_{\hat{\theta}}(y) + \frac{1}{2} u(\theta_0)^T H^{-1}(\theta_0) u(\theta_0)\end{aligned}$$

$\hat{\theta} \xrightarrow{P} \theta_0$

For large n , $H(\theta_0) \approx \mathbb{E}_Y[H(\theta_0)] = -A$

$$\begin{aligned}\text{so: } \mathbb{E}_Y[-2 \log f_{\theta_0}(y)] &\approx \mathbb{E}_Y[-2 \log f_{\hat{\theta}}(y)] + \\ &\quad \mathbb{E}_Y[u(\theta_0)^T A^{-1} u(\theta_0)] \\ &= \mathbb{E}_Y[-2 \log f_{\hat{\theta}}(y)] + \text{tr}(A^{-1}B)\end{aligned}$$

Trace trick: $\mathbb{E}[x^T C x] = \mathbb{E}[x]^T C \mathbb{E}[x] + \text{tr}(C \text{Var}(X))$

AIC

Akaike information criterion

Let $\hat{h}$ be a model fit on training data $\mathcal{T}$ by maximizing likelihood (equivalently, minimizing deviance). Let p be the number of parameters in the model (i.e. for logistic regression, $\beta \in \mathbb{R}^p$).

$$AIC(\hat{h}) = \text{deviance} + \underbrace{(2p)}_{\text{penalty on \# of parameters}}$$

($\Rightarrow$ if model is correctly specified, $\mathcal{R} \approx \mathbb{E}[\frac{1}{n} AIC]$)

Example

```
m_dengue <- glm(Dengue ~ Age + WBC + PLT,  
                 data = dengue, family = binomial)  
m_dengue$deviance
```

```
## [1] 5200.823
```

$$5200.823 + 2(4) = 5208.823$$

```
m_dengue$aic
```

```
## [1] 5208.823
```